Thermodynamic description of the Al–Cu system: a synthetic benchmark document

ORCHID synthetic benchmark (generated; no human authors)

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Abstract

This synthetic document describes the Al–Cu binary system as calculated with pycalphad from the SGTE dataset AlCu-02Mie.tdb; every figure, table and stated value is generated, and the exact values are kept in a separate truth key. The highest invariant reaction, L → β + (Cu), lies at 1036.7 °C.

1. Introduction

The Al–Cu system belongs to the 20 binaries of the ORCHID benchmark, chosen to cover diagram types such as intermetallic rich, peritectic. The thermodynamic description is the SGTE collection entry (Hallstedt, 2025), which reproduces the assessment of J. Miettinen, Metall. Mater. Trans. A, 33A, 1639-48(2002). The calculations use pycalphad 0.11.2 at 1 atm.

For Al, the liquidus meets the axis at 660.3 °C. The melting point of Cu reproduced by the dataset is 1084.6 °C.

2. Phase diagram

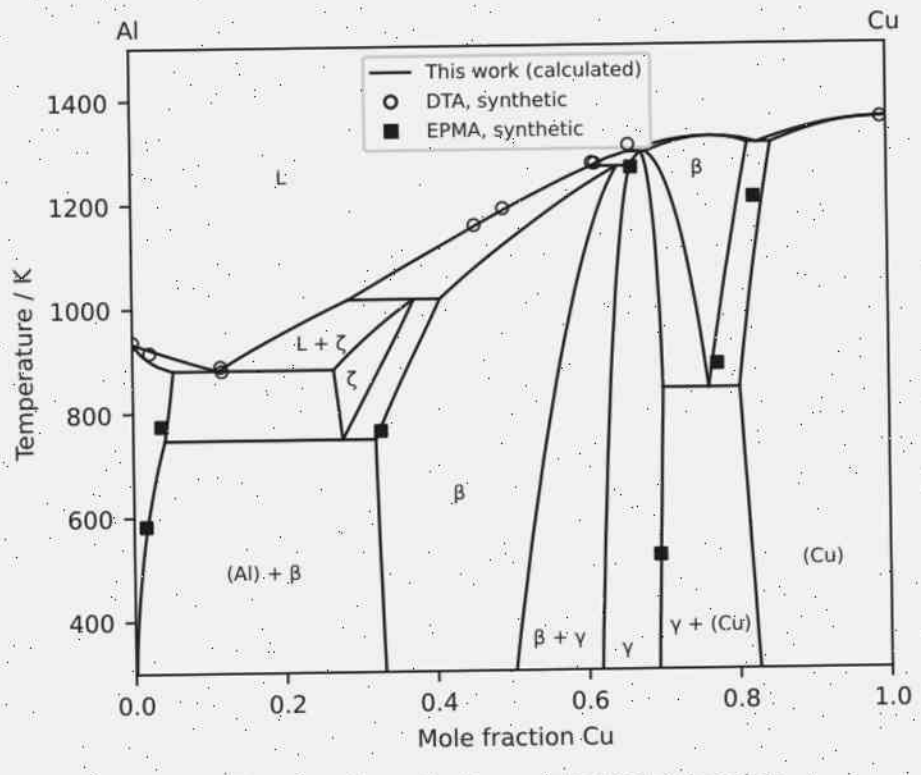


Fig. 1. Calculated Al–Cu phase diagram with synthetic experimental-style points.

The eutectic reaction L → β + (Cu) is calculated at 1036.7 °C, with L at a mole fraction of 0.829. At 1020.6 °C the peritectic L + β → γ takes place; the L phase then contains x(Cu) = 0.658. Peritectic equilibrium L + γ → β is found at 993.5 °C and x(Cu) = 0.609 (L). The dataset places a peritectic (L + β → ζ) at 742.1 °C; the composition of L is a mole fraction of 0.286. The eutectic reaction L → (Al) + ζ is calculated at 607.9 °C, with L at x(Cu) = 0.113. Eutectoid equilibrium β → γ + (Cu) is found at

566.9 °C and x(Cu) = 0.762 (β). The eutectoid reaction ζ → (Al) + β is calculated at 473.9 °C, with ζ at a mole fraction of 0.277. β melts congruently at 1049.9 °C and x(Cu) = 0.767.

Table 1. Invariant reactions in the Al–Cu system (calculated).

Reaction	Type	T (K)	at.% Cu, 1	at.% Cu, 2	at.% Cu, 3
L → β + (Cu)	eutectic	1309.9	82.9	81.6	84.7
L + β → γ	peritectic	1293.8	65.8	68.2	67.5
L + γ → β	peritectic	1266.7	60.9	66.1	64.4
L + β → ζ	peritectic	1015.2	28.6	40.7	37.3
L → (Al) + ζ	eutectic	881.1	11.3	5.4	26.6
β → γ + (Cu)	eutectoid	840.0	76.2	70.1	80.3
ζ → (Al) + β	eutectoid	747.0	27.7	4.2	32.1

Phases 1–3 in the order written in the reaction. Calculated with pycalphad 0.11.2 from the SGTE dataset; synthetic document.

3. Thermodynamic properties

At 1450 K the integral mixing enthalpy of the liquid has its minimum, −17.57 kJ/mol, at x(Cu) = 0.600. The calculated enthalpy of formation of ζ at 298.15 K is −11.10 kJ/mol per mole of atoms.

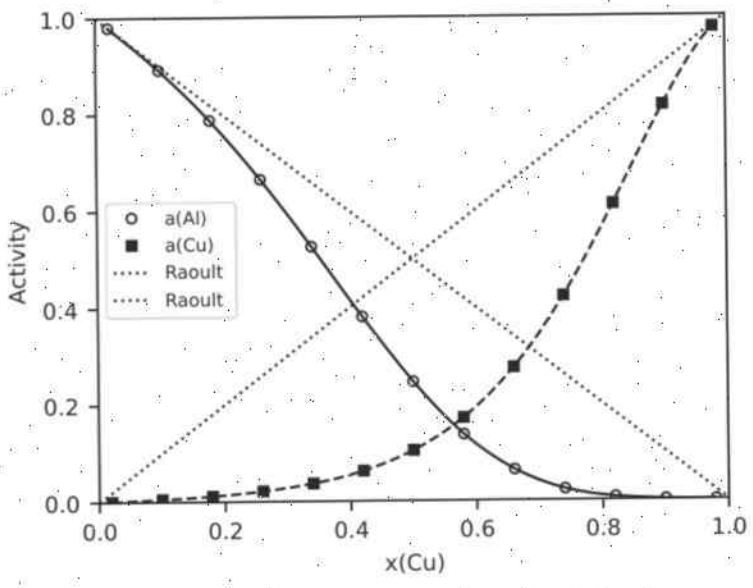


Fig. 2. Calculated activities in the liquid at 1450 K (reference: pure liquids).

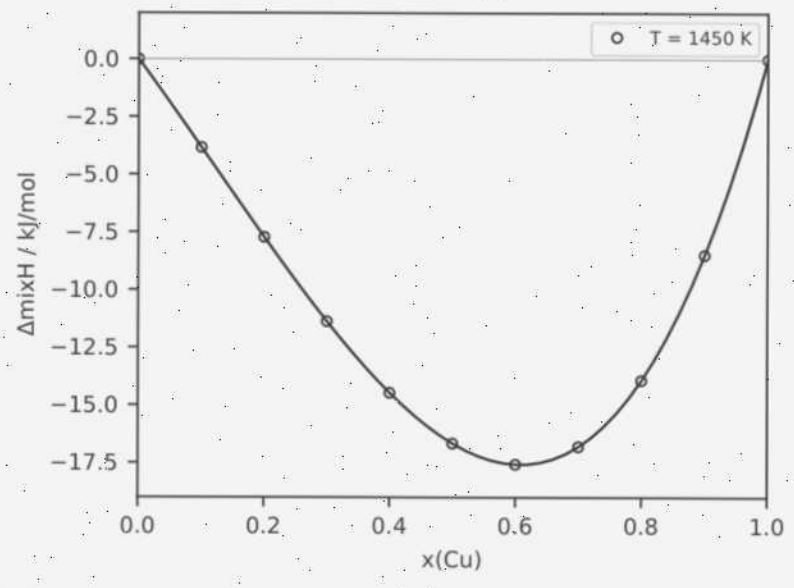


Fig. 3. Calculated mixing enthalpy of the liquid at 1450 K.

Table 2. Calculated formation properties of the intermediate phases (Al–Cu).

Phase	at.% Cu	ΔfH(298 K) / kJ mol⁻¹	S(298 K) / J mol⁻¹ K⁻¹
γ	66.6	−20.31 ± 1.0	32.80
β	46.9	−18.66 ± 0.5	32.31
ζ	28.9	−11.10 ± 1.5	32.84

Reference: stable elements (SER) at 298.15 K. ± values are synthetic illustration uncertainties, not data.

4. Conclusions

This document is synthetic and exists only to test extraction tools; its truth key lists every boundary, field, invariant, plotted point, table cell and stated value.